PS Response: Problem Set 1

Applied Stats/Quant Methods 1

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Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```r
y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
    80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school.

```r
mean_y <- mean(y) # sample mean
n <- length(y)    # sample size == 25
s <- sd(y)        # sample standard deviation
SE <- s/sqrt(n)   # standard error of the mean
alpha <- 0.10
df<- n-1          # degrees of freedom == 24
tcrit_1 <- qt(1 - alpha/2, df) #two-sided 90% CI -> qt(0.95.24)
tcrit_1           # == 1.710882

ci_90_lower <- mean_y - tcrit_1 * SE
ci_90_upper <- mean_y + tcrit_1 * SE
```

```
> print(mean_y)
[1] 98.44
> print(n)
[1] 25
> print(s)
[1] 13.09339
```

```
> print(SE)
[1] 2.618556
> print(tcrit)
[1] 1.710882
> print(ci_90_lower)
[1] 93.95993
> print(ci_90_upper)
[1] 102.9201
```

The sample mean was 98.44 with $n = 25$, $s = 13.09$, and standard error 2.62. Using a t-distribution with $df = 24$ and $t_{0.95,24} = 1.711$, the 90% confidence interval for the average IQ in the school is:

$$(93.96,\ 102.92).$$

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

   Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

```

mu0 <- 100          # H0: mu=100, H1: mu > 100
t_stat <- (mean_y - mu0) / SE
t_crit <- qt(0.95, df) # because alpha is 0.05
```

```
print(n)
[1] 25
print(mean_y)
[1] 98.44
print(s)
[1] 13.09287
print(SE)
[1] 2.618575
print(t_stat)
[1] -0.5957439
print(t_crit)
[1] 1.710882
print(p_value)
[1] 0.7215383
```

I conducted a one-sided t-test with $H_0 : \mu = 100$ and $H_1 : \mu > 100$. The sample mean was 98.44 with $n = 25$, $s = 13.09$, and standard error 2.62. The test statistic was

$t = -0.596$, the critical value at $\alpha = 0.05$ was 1.711, and the one-sided p-value was 0.72. Since $p > 0.05$, we fail to reject the null hypothesis and conclude that there is no statistical evidence that the school mean IQ is greater than 100.

Or, I can simply use ***t.test*** function to get an answer.

```
t.test(y, mu = 100, alternative = "greater")
```

```
One Sample t-test

data:  y
t = -0.59574, df = 24, p-value = 0.7215
alternative hypothesis: true mean is greater than 100
95 percent confidence interval:
93.95993      Inf
sample estimates:
mean of x
98.44
```

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

`State`	*50 states in US*
`Y`	*per capita expenditure on shelters/housing assistance in state*
`X1`	*per capita personal income in state*
`X2`	*Number of residents per 100,000 that are "financially insecure" in state*
`X3`	*Number of people per thousand residing in urban areas in state*
`Region`	*1=Northeast, 2= North Central, 3= South, 4=West*

Explore the `expenditure` data set and import data into `R`.

- Please plot the relationships among *Y*, *X1*, *X2*, and *X3*? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```r

# Keep only Y, X1, X2, X3
vars <- expenditure[, c("Y", "X1", "X2", "X3")]

# Pairwise scatterplot matrix (base R)
pdf("q2_pairs.pdf", width = 7, height = 7)  # save to pdf
pairs(vars,
      main = "Pairwise Plots: Y, X1, X2, X3")  # simple, transparent
dev.off()

# Focused scatterplots of Y vs each X (side by side)
pdf("q2_Y_vs_Xs.pdf", width = 9, height = 3)  # save to pdf
op <- par(mfrow = c(1, 3))  # three panels in one row

plot(vars$X1, vars$Y,
     xlab = "X1: Per-capita personal income",
     ylab = "Y: Per-capita expenditure",
     main = "Y vs X1",
     pch = 19, col = "lightpink")

plot(vars$X2, vars$Y,
     xlab = "X2: Financially insecure per 100,000",
     ylab = "Y: Per-capita expenditure",
     main = "Y vs X2",
     pch = 19, col = "skyblue")

plot(vars$X3, vars$Y,
     xlab = "X3: Urban residents per 1,000",
     ylab = "Y: Per-capita expenditure",
     main = "Y vs X3",
     pch = 19, col = "lightgreen")

par(op)  # restore graphics setting
dev.off()

# Correlation matrix (Pearson, base R)
cor_mat <- cor(vars, use = "pairwise.complete.obs", method = "pearson")
cor_mat            # raw numbers
round(cor_mat, 3)  # rounded to 3 decimals

# Print each correlation explicitly
cat("corr(Y, X1) =", round(cor_mat["Y","X1"], 3), "\n")
```

```
> round(cor_mat, 3)
Y    X1    X2    X3
Y   1.000 0.532 0.448 0.464
X1  0.532 1.000 0.206 0.595
X2  0.448 0.206 1.000 0.221
X3  0.464 0.595 0.221 1.000
```

```
> corr(Y, X1) = 0.532
> corr(Y, X2) = 0.448
> corr(Y, X3) = 0.464
```

Figure 1: Pairwise scatterplots among Y, X_1, X_2, and X_3.

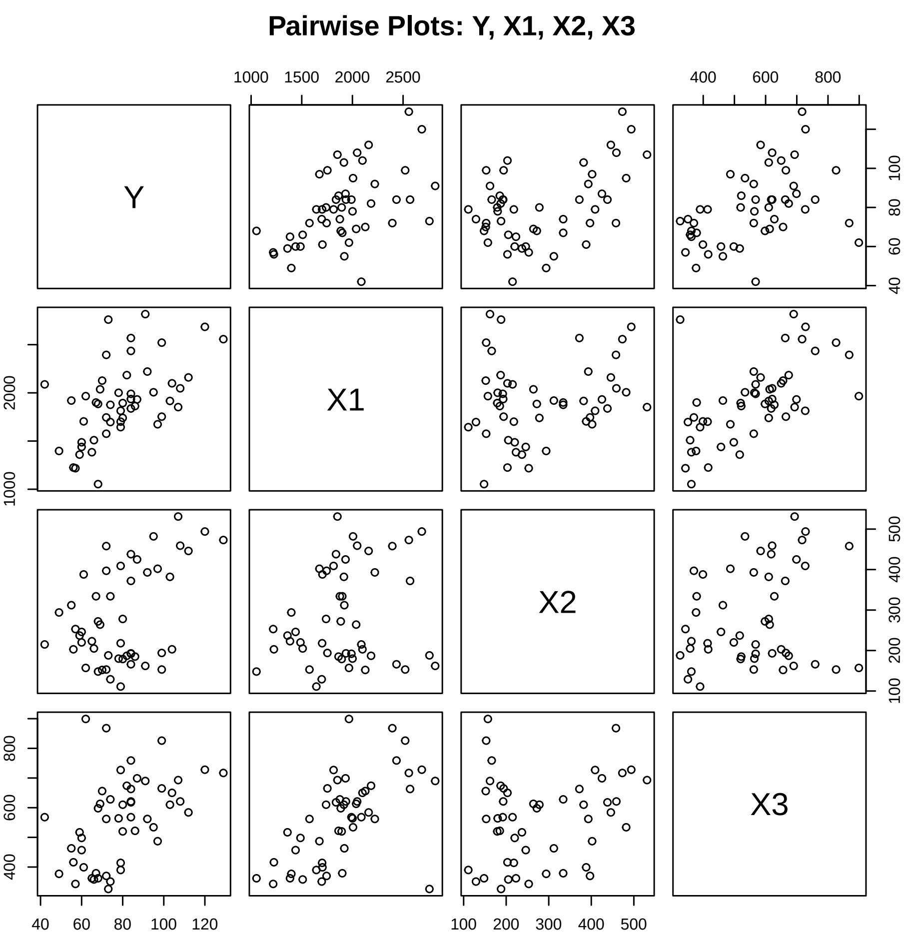
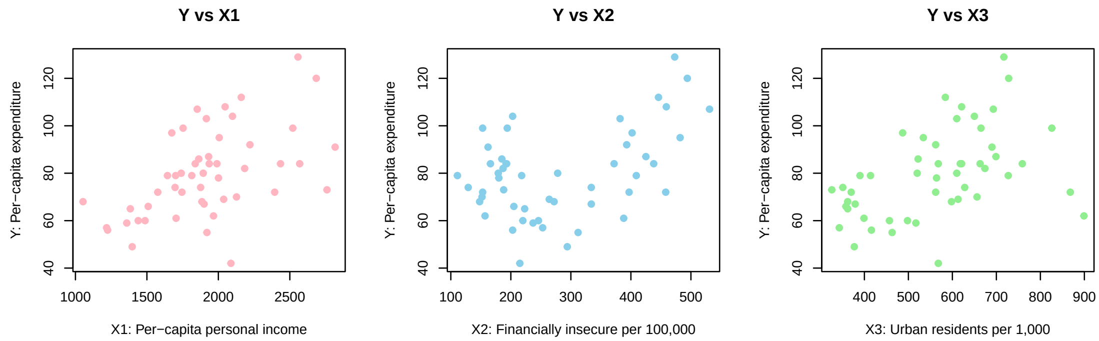


Figure 2: Scatterplots of Y vs X_1, X_2, and X_3.



I first used a simple **pairwise scatterplot matrix** (`pairs`) because it shows every bivariate relationship among (Y, X_1, X_2, X_3) at once, making it easy to screen for linear trends, direction (positive/negative), and potential outliers without relying on advanced libraries. I then added three focused scatter plots of Y against each of X_1, X_2, X_3 to describe the substantive relationships more clearly in the text.

The Pearson correlation matrix provides numerical support for the visual impressions. In particular, $\mathrm{corr}(Y, X_1) = 0.532$, $\mathrm{corr}(Y, X_2) = 0.448$, and $\mathrm{corr}(Y, X_3) = 0.464$ (fill in from the R output). Based on these values and the scatter plots, I describe the relationships in terms of direction (positive/negative), approximate strength (weak/-moderate/strong), and any visible nonlinearity or outliers.

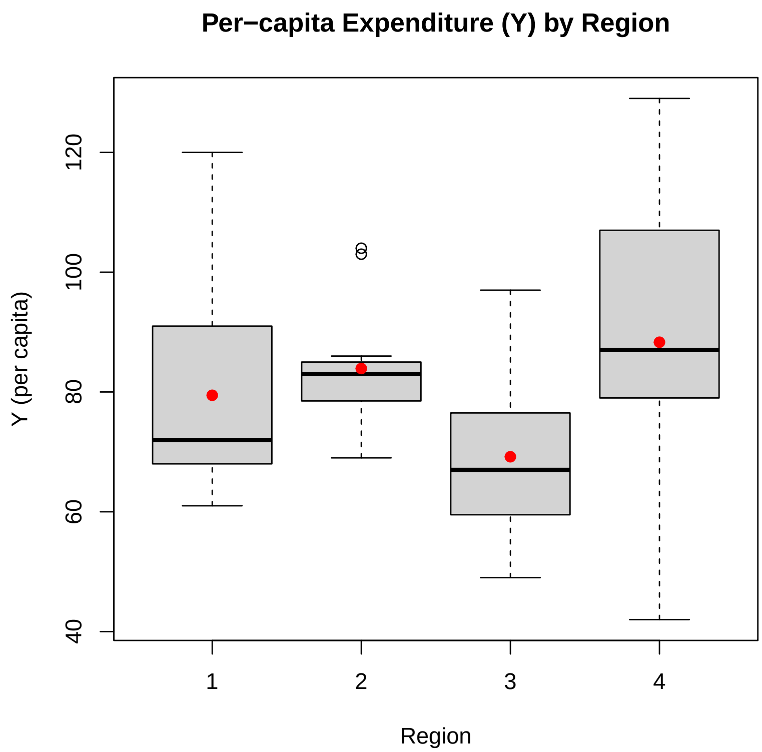
- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```
pdf("q2_Y_by_region.pdf", width = 6, height = 6)
boxplot(Y ~ Region, data = expenditure,
        main = "Per-capita Expenditure (Y) by Region",
        xlab = "Region", ylab = "Y (per capita)")
region_means <- tapply(expenditure$Y, expenditure$Region, mean, na.rm=
    TRUE)
points(1:length(region_means), region_means, col="red", pch=19)
dev.off()

# Print means per Region
print(region_means)
```

```
> print(region_means)
1         2         3         4
106.7500  85.7083   77.2400   97.5455
```

Figure 3: Boxplot of Y by Region, red dots indicate region means.



I chose a **boxplot** because it clearly shows the distribution of Y within each region (spread, median, outliers) and allows easy comparison across groups. Adding red dots for region means makes it easy to identify which region spends the most. From the output, Region 1 (Northeast) has the highest mean per-capita expenditure on housing assistance.

- Please plot the relationship between Y and *X1*? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

- Please plot the relationship between Y and *X1*? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```
pdf("q2_Y_vs_X1.pdf", width = 6, height = 6)
plot(expenditure$X1, expenditure$Y,
     main = "Y vs X1",
     xlab = "Per-capita personal income (X1)",
     ylab = "Per-capita expenditure (Y)",
```

```r
      pch = 19, col = "gray40")
abline(lm(Y ~ X1, data = expenditure), lty = 2, lwd = 2)
dev.off()

# Scatterplot Y vs X1 by Region
pdf("q2_Y_vs_X1_by_region.pdf", width = 6.5, height = 6)
cols <- c("blue", "orange", "green", "purple")
pchv <- c(16, 17, 15, 18)
plot(expenditure$X1, expenditure$Y,
      main = "Y vs X1 by Region",
      xlab = "Per-capita personal income (X1)",
      ylab = "Per-capita expenditure (Y)",
      col = cols[as.integer(expenditure$Region)],
      pch = pchv[as.integer(expenditure$Region)])
abline(lm(Y ~ X1, data = expenditure), lty = 2, lwd = 2)
legend("bottomright",
        legend = levels(factor(expenditure$Region)),
        col = cols, pch = pchv, bty = "n",
        title = "Region")
dev.off()
```

Figure 4: Scatterplot of Y vs X_1 with fitted regression line.

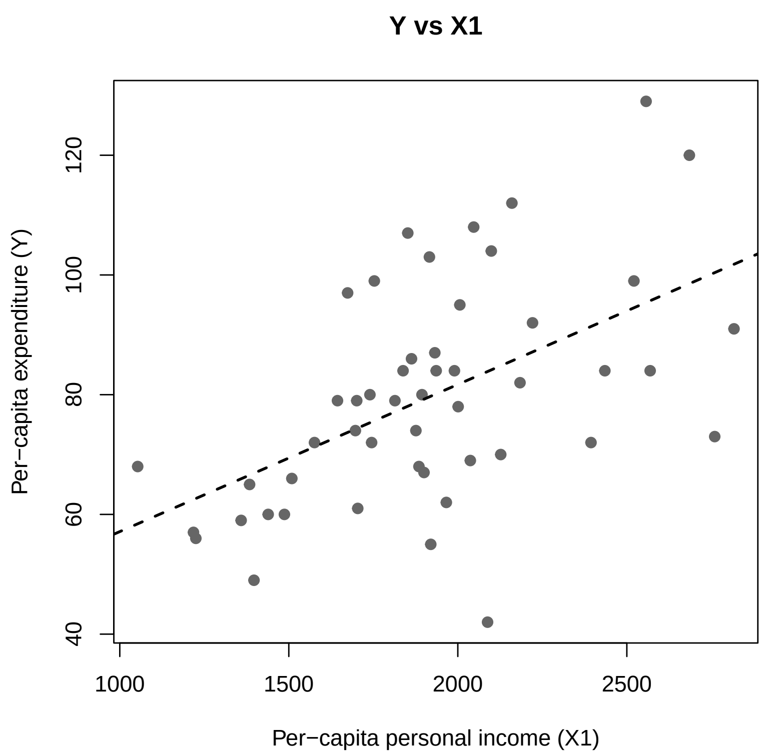
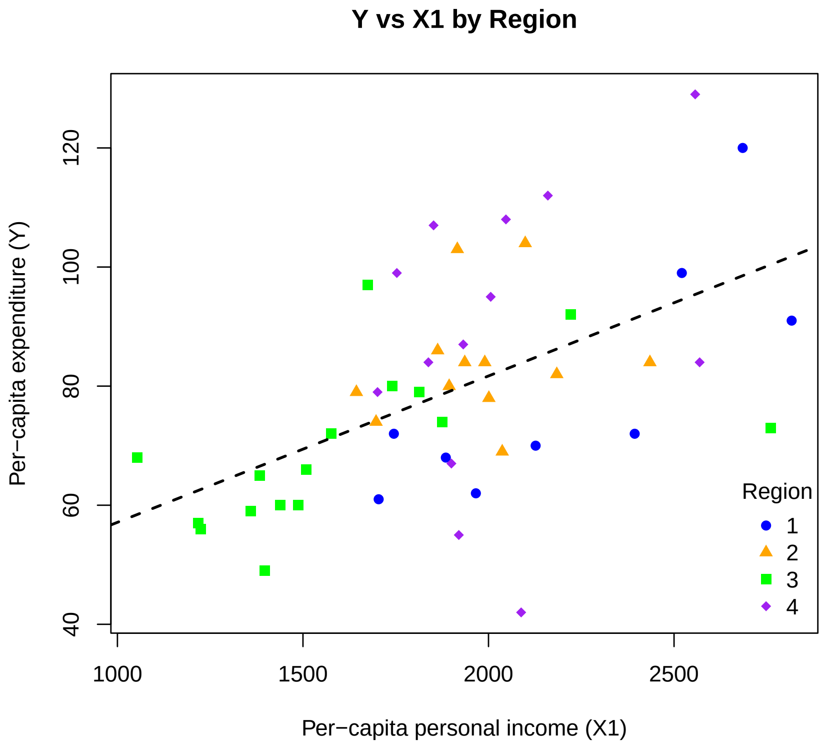


Figure 5: Scatterplot of Y vs X_1 by Region. Colors and symbols distinguish regions, dashed line shows overall linear trend.



The scatterplot of Y versus X_1 suggests a moderate positive relationship: states with higher per-capita income tend to spend more on housing assistance. When we add *Region*, the points are distinguished by color and symbol, revealing that while the overall trend is positive, regional differences are also present (e.g., Region 1 states often cluster toward higher spending).